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10 September 1976
SUPERSEDING
MIL-P-14835A(MU)
5 May 1972

MILITARY SPECIFICATION

PROJECTILE, SMOKE, WP, 57MM: M308A1 METAL PARTS ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers Projectiles, Smoke, WP, 57MM: M308A1 Metal Parts Assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition, General Specification for
MIL-M-6867	- Magnetic Inspection Units
MIL-I-6868	- Inspection Process Magnetic Particle

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-1167	- Ammunition Data Card
MIL-STD-1168	- Lot Numbering of Ammunition
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235	- Single and Multilevel Continuous Sampling Procedure and Tables for Inspection by Attributes

FSC 1310

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in approving this document should be addressed to: Commander, Frankford Arsenal, ATTN: SARFA-MDM, Phila., PA 19137 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.1 (Cont'd)

DRAWINGS

Armament Command

D10543043	- Projectile, Smoke, WP: 57mm, M308A1, Metal Parts Assembly
I.I-10543043	- Inspection Equipments List for Projectile, Smoke, WP, 57mm, M308A1
F7548319	- Box Fiberboard Ammunition, For Interplant Shipment of Projectiles Empty

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ANSI-B46.1	- Surface Texture

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, NY 10018.)

ASTM E-8	- Tension Testing of Metallic Materials
ASTM E-10	- Brinell Hardness of Metallic Materials
ASTM B-117	- Salt Spray (Fog) Testing

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Phila., PA 19103.)

3. REQUIREMENTS

3.1 General.-Material, parts and assembly shall comply with the requirements specified on the applicable drawing, referenced specifications, and the following:

3.2 Mechanical properties.-The mechanical properties required by the applicable drawing shall be uniform throughout each component.

3.2.1 Heat treatment or stress relief.-Heat treatment or stress relief applied to any component part shall be uniform throughout and shall be so conducted as to produce the mechanical properties specified on the applicable drawing.

3.4 Protective coating.-Requirements relative to paint thickness are not specified, except as a guide for performance of tests as indicated by the applicable drawings and specifications. Actual body assemblies must withstand the specified salt-spray exposure with coating breakdown no greater than that described by the applicable specification.

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3.5 Press-fit surface.-The small inside diameter of the Adapter is a critical press-fit surface and must remain free of dirt and products of corrosion, and must remain undamaged for its entire length.

3.6 Metal defects.

3.6.1 Components (body and adapter).-The components, including the cavity, shall be free from cracks, splits, burst, cold shuts, pipe, porosity, inclusions, folds, seams and other metal defects.

3.6.2 Components (body and adapter).-The cavity of the components shall be free of scale, fins, burrs, draw marks, laminations, imbedded foreign matter, pits and sharp edges.

3.6.3 Visual standards.-Visual standards for metal defects of paragraph 3.6.2 will be established for each contract after the first month's production, or whenever requested by the procuring activity (see 6.2).

3.7 Proving ground functional test.-Projectiles, when tested at excess and service pressure, shall show no evidence of loss or break up of metal parts components in flight.

3.8 Workmanship.-The requirements for workmanship are shown on the applicable drawing, in the applicable specification and the following:

3.8.1 Threads.-Threads shall be full and undamaged for the entire minimum depth as required by the applicable drawing and shall be in accordance with the provisions of MIL-A-2550.

3.8.2 Surface finish.-The requirements of surface finish are as detailed on the applicable drawing and in the applicable specifications.

3.8.3 Burr.-No part shall have a burr which might interfere with the assembly of the round or which might be injurous to personnel handling the item.

3.8.4 Foreign matter.-No parts or assmebly shall contain dirt, grease, chips, rust, corrosion or other foreign matter.

3.8.5 Brazed joint.-The brazed metal shall be visible around the entire area (360 degrees) where the adapter flange joins the body wall.

3.8.6 Joint quality.-When sectionalized and peeled, the brazed joint shall be free of entrapped slag, voids, unwetted or unbonded areas and other brazing defects. The aggregate area of these defects shall not exceed 15 percent of the area of the faying surfaces. The maximum extent of the defect shall not exceed 15 percent of the overlap distance (width) of the joint. Presence of cleaning or other fluids in the solder groove shall be considered as evidence of a defective joint.

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3.8.7 Rotating band.-After machining of the rotating band, care shall be taken to protect it from being nicked, burred or otherwise damaged during subsequent handling and preparation for shipment.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.-Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection.-Inspection shall be in accordance with MIL-A-2550, except as specified herein.

4.1.3 Inspection criteria.

a. Successful passage of all inspection required by the drawings and specifications shall be considered as a condition precedent to performing the Proving Ground functional test.

b. Acceptance or rejection of a lot by the Government shall be contingent on successful passage of the Proving Ground functional tests.

4.2 First article sample.

4.2.1 Initial production sample.-At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 70 metal parts assemblies as follows:

55 Metal Parts Assemblies, complete, painted
5 Metal Parts Assemblies, unpainted
10 Sets of component parts, unassembled

Fifty of the above painted metal parts assemblies shall be forwarded to a designated proving ground for proving ground functional tests. The sample shall be manufactured, using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply. A first article sample or portion thereof as directed by the procuring activity shall also be

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4.2.1 (Cont'd)

submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing processes or procedures, material used, drawing, specifications or source of supply such as to significantly affect product uniformity as determined by the responsible technical agency.

4.2.1.1 Examination and test.-After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure.-Failure of the sample to comply with requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.-The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Heat treatment lots.-Heat treatment lots shall contain component forgings from not more than one heat of steel that have been heat treated in the same heat treating equipment, at the same temperature, in one unchanged process. The heat treatment lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

4.3.1.3 Stress relieved lots.-Stress relieved lots shall contain cold worked components from not more than one heat of steel that have been cold worked by the same process, utilizing the same tooling and equipment and stress relieved in the same furnace at the same temperature, at the same length of time. The stress relieved lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

4.3.1.4 Bar stock lot.-Components machined from bar stock shall contain material from not more than one heat of steel and produced in one unchanged process. Wherever heat treatment is applicable (whether during manufacture of bar stock or manufacture of component, the heat treatment shall be performed in the same heat treating equipment at the same temperature in one unchanged process. The machining lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

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4.3.1.5 Miscellaneous lots.-Small lots (such as bodies or adapter from abnormally small heats of steel or bodies or adapters that have become separated from previously delivered lot, etc.) may at the discretion of the supplier be grouped into miscellaneous heat treatment, stress relieved or machining lots as the case may be. Such lots shall be identified by a miscellaneous lot code number and processed in accordance with either 4.3.2.1, 4.3.1.3, or 4.3.1.4, as applicable. The miscellaneous lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

4.3.1.6 Projectile lots.-When it has been determined that bodies of a heat treated or stress relieved lot have complied with the mechanical properties requirement of the applicable drawing, they may be grouped for further processing.

4.3.1.7 Shipping lots.-Shipping lots shall consist of projectile assemblies produced by one supplier, under one contract, in one unchanged process, in accordance with the same drawings, revisions, under conditions of a continuous production, free of interruptions other than those due to the end of shift day, or work week. The shipping lot thus constituted shall be the basis for selection of samples for the proving ground functional test. Lot size shall be determined by the procuring contracting officer.

4.3.2 Examination.-One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans contained in MIL-STD-105, continuous sampling plans in accordance with standard MIL-STD-1235 may be used if approved by the procuring activity. An AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect shall be used. All nonconforming material shall be rejected. The roughness comparison specimens prescribed by ANSI-B46.1 shall be used as a basis for surface roughness determination except press-fit surface. Surface finsih of the press-fit surface shall not be inspected until all manufacturing and gaging operations have been completed. The miniature inspection lamp, listed on the LI, shall be used for examination to determine tangency of the radius at the mouth end of the press-fit surface. Four power (4x) magnification may be used in performing visual examination.

4.3.2.1 Tool control dimensions.-

a. Dimensions marked tool control shall be gaged at the beginning or end of each working day (24 hrs.), during First Article Inspection and for the first component after each tooling change, adjustment or repair. Where destruction of the component is necessary to gage such dimensions, measurements of the tool may be substituted provided the contractor has established correlation between the tool dimensions and the component dimensions prior to the start of production.

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b. When the contractor has satisfied (a) above, visual inspection of tool control dimensions will be permitted. Visual inspection shall be performed every two (2) hours for each machine in operation. If any tool control dimension(s) is in doubt during visual inspection, the dimension(s) shall be immediately gaged (see "a" above) to determine acceptability. Corrective action will immediately be instituted by the contractor whenever a defective tool control dimension is found.

4.3.2.2 Adapter, Drawing C10543047, covering a detail of Drawing D10543043.

Defect	Method of Inspection
CRITICAL	
1. Metal defective	Visual
MAJOR	
101. Length of brazing surface	Gage
MINOR	
201. Diameter of shoulder	Gage
202. Diameter of boss at rear	Gage
203. Depth of solder ring groove, max. (if applicable)	Gage
204. Length from front face to shoulder	Gage
205. Length of flat on shoulder, max.	Gage
206. Length, total	Gage
207. Location of solder ring groove	Gage
208. Width of solder ring groove, max.	Gage
209. Radii or chamfers, missing or incorrect	Visual (See 6.1)
210. Surface finish improper	Visual (See 6.1)
211. Burr	Visual
212. Foreign matter	Visual

4.3.2.3 Body, Drawing F10543046, covering a detail of Drawing D10543043.

CRITICAL	
1. Metal defective	Visual (Probe Light)
MAJOR	
101. Runout of charge cavity with bourrelets	Gage
102. Diameter of brazing cavity	Gage

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4.3.2.3 (Cont'd)

Defect	Method of Inspection
103. Diameter of charge cavity	Gage
104. Diameter of forward end	Gage
105. Length from front face to rear of forward bourrelet	Gage
106. Thickness of wall, min.	Gage
107. Thickness through base	Gage

MINOR

201. Diameter of chamfer at brazing surface	Gage
202. Length, total	Gage

4.3.2.4 Metal Parts Assembly, Drawing D10543043 (prior to paint and insertion of protector cap).

CRITICAL

1. Braze metal not visible for full 360 degrees at where adapter joins body wall	Visual
2. Air test stamp missing (indicating failure to pass air test)	Visual
3. Metal defective	Visual (Probe Light)

MAJOR

101. Weight	Approved Scale
102. Runout of pitch diameter of thread with bourrelets and front face	Gage
103. Runout of press fit surface with bourrelets	Gage
104. Runout of body with bourrelets	Gage
105. Runout rear of rotating band with bourrelets	Gage
106. Runout of rotating band lands with bourrelets	Gage
107. True position of press fit surface counterbase with press fit diameter	Gage
108. True position of rotating band grooves with rotating band lands	Gage
109. Diameter of bourrelets, min.	Gage
110. Diameter of body	Gage
111. Diameter of rotating band lands	Gage
112. Diameter of rotating band grooves	Gage
113. Diameter of press fit surface counterbore	Gage
114. Depth of crimp grooves	Gage
115. Length from rear face to rear of rotating band	Gage
116. Length of rotating band	Gage
117. Length from front of rotating band to front of rear bourrelet	Gage

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4.3.2.4 (Cont'd)

Defect	Method of Inspection
118. Length from rear of rotating band to front of rear rotating band relief groove	Gage
119. Width of rotating band lands	Gage
120. Width of rotating band grooves	Gage
MINOR	
201. Diameter of undercut on rotating band lands, min. (if applicable)	Gage
202. Depth of indentations on rear face, max.	Gage
203. Length to undercuts on rotating band lands, min. (if applicable)	Gage
204. Depth of undercuts, front and rear of rotating band (if applicable)	Visual (See 6.1)
205. Width of undercuts front and rear of rotating band (if applicable)	Visual (See 6.1)
206. Radii of chamfers, missing or incorrect	Visual (See 6.1)
207. Surface finish, improper	Visual (See 6.1)
208. Burr	Visual
209. Foreign matter	Visual

Special inspection for projectile assembly (prior to paint)

Defect	Method of Inspection
(a) Diameter of press-fit surface	Gage
(b) Reverse taper of press-fit surface	Gage
(c) Surface finish of press-fit surface	Gage
(d) Radius not tangent to press-fit surface for full 360 degrees	(See 4.3.2)

Inspection for the above defect characteristics shall be performed on an individual basis as follows:

(1) One hundred percent inspection will be in effect for each individual characteristic until the characteristic is found to be defect free after 2500 consecutive inspections, then that characteristic may qualify for reduced inspection of 1 in 5.

(2) Should a characteristic be rejected while on reduced inspection, that characteristic will be backscreened until 15 consecutive projectiles are found free of the rejected characteristic and in addition, revert back to (1) above.

(3) For requalification of reduced inspection for any characteristic, the contractor must follow the procedures outlined in paragraph (1) above.

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4.3.2.5 Metal parts assembly, Drawing D10543043 (after paint and prior to insertion of protector cap).

CRITICAL		
1.	Metal defective	Visual (Probe light)
MAJOR		
101.	Pitch diameter of thread, max.	Gage
102.	Minor diameter of thread, max.	Gage
103.	Pitch diameter of thread, min.	Gage
104.	Diameter of bourrelets, max.	Gage
105.	Depth to bottom of thread cavity	Gage
106.	Depth to bottom of press-fit cavity diameter counterbore	Gage
107.	Length of effective thread, min.	Gage
108.	Profile of rotating band, max. (functional)	Gage
109.	Form of radius at press-fit surface improper	Visual (See 6.1)
110.	Rotating band damaged	Visual
111.	Crystalline or Tablet VTI, missing	Visual
MINOR		
201.	Diameter of thread chamfer	Gage
202.	Diameter of nose	Gage
203.	Protective coating inadequate or damaged	Visual
204.	Paint on thread(s)	Visual
205.	Paint on front face	Visual
206.	Paint not dry	Visual/Tactile
207.	Corrosion preventive compound inadequate or excessive (indicated areas)	Visual
208.	Marking missing, incorrect of illegible	Visual
209.	Protector cap, missing	Visual

4.3.2.6 Packed carton (prior to sealing).

CRITICAL		
	None defined	
MAJOR		
	None defined	
MINOR		
201.	Support missing or damaged	Visual
202.	Spearator missing or damaged	Visual
203.	Improper assembly	Visual
204.	Incorrect number in carton	Visual

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4.3.2.7 Packed carton (after sealing).

CRITICAL

None defined

MAJOR

None defined

MINOR

201.	Carton damaged, cut or gouged	Visual
202.	Carton not completely sealed	Visual
203.	Contents loose	Visual/Manual
204.	Marking missing, incorrect or illegible	Visual

4.3.3 Inspection testing.-The following tests shall be performed in accordance with provisions of 4.4:

4.3.3.1 Mechanical properties.(See 6.3)

4.3.3.1.1 Hardness.

4.3.3.1.1.1 Heat treatment.-In order to determine uniformity of heat treatment, each body forging in the heat treatment lot (as defined by 4.3.1.2) shall be subjected to a hardness test with a Brinell hardness tester, or by any other method approved by the procuring activity.

4.3.3.1.1.2 Stress relief lots.-Twenty-five stress relieved bodies from the first batch of each stress relief lot (as defined by 4.3.1.3) processed through thr stress relief furnace and five bodies from each hour's production thereafter for the balance of the lot shall be subjected to a hardness test with a Brinell hardness tester or by any other method approved by the procuring activity, to assess uniformity of the stress relief furnace. Hardness values shall be recorded.

4.3.3.1.1.3 Bar stock lot.-When mechanical properties of the bar stock are altered by heat treatment during the manufacture of components, each component in the bar stock lot (as defined by 4.3.1.4) shall be subjected to a hardness test with a Brinell hardness tester or by any other method approved by the procuring activity in order to determine uniformity of heat treatment.

4.3.3.1.1.4 Miscellaneous lots.-Miscellaneous heat treatment lot shall be subjected to a hardness test in accordance with 4.3.3.1.1.1. Miscellaneous stress relieved lot shall be subjected to a hardness test in accordance with 4.3.3.1.1.2. Miscellaneous machined bodies lots shall be subjected to a hardness test in accordance with 4.3.3.1.1.3.

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4.3.3.1.2 Tension test.

4.3.3.1.2.1 Heat treatment lots.-Two projectile bosied, one of the hardest and one of the softest as determined by 4.3.3.1.1.1 or 4.3.3.1.1.3 shall be subjected to this test. Failure of the tension test specimen to comply with the yeild strength or elongation requirements of the applicable drawing shall be cause for rejection of the lot. At the option of the supplier, a production control sample may be utilized to establish a hardness range. When a production control sample is used, the hardness thus established shall be maintained throughout the heat treatment lot. The size of the sample shall be determined by the contractor. Projectile bodies which fall outside of the established range may be re-heat treated to fall within the range or new mechanical properties may be determined from newly selected projectile bodies which will represent the true extremes of hardness on the heat treatment lot.

4.3.3.1.2.2 Stress relieved lots.-Two bodies, one of the hardest and one of the softest as determined by the hardness test from the first twenty-five bodies selected in accordance with 4.3.3.1.1.2 shall be subjected to this test. Failure of any tension test specimen to comply with the mechanical properties requirements of the applicable drawing shall be cause for rejection of the lot. When it has been determined that the mechanical properties comply and furnace temperature and time have been established, the furnace time and temperature shall be maintained throughout the stress relieved lot. The furnace temperature may vary within twenty-five degrees F., but not below the minimum temperature of 700 degrees F., and the soak time of 30 minutes minimum. Furnace temperature and time will be recorded every hour to assess adequate control of stress relief equipment. The furnace temperature will be calibrated every eight hours by an independent method.

4.3.3.1.2.3 Bar stock lots.-When heat treatment is not required during manufacture of components from bar stock; two projectiles from each lot of 5000 projectiles or fraction thereof shall be subjected to this test failure of any tension test specimen to comply with the yield strength or elongation requirements of the applicable drawing shall be cause for rejection of the production represented by the sample.

4.3.3.1.2.4 Miscellaneous lots.-Samples for tension test of miscellaneous heat treatment, stress relieved or bar stock lots shall be selected in accordance with 4.3.3.1.2.1, 4.3.3.1.2.2, 4.3.3.1.2.3, or 4.3.3.1.2.5 as applicable. Failure of any tension test specimen to comply with the yield strength or elongation requirements of the applicable drawing shall be cause for rejection of the lot. Double the number of samples shall be selected for miscellaneous lots.

4.3.3.1.2.5 Adapters.-Two adapters from each heat treated, or stress relieved lot shall be selected and subjected to this test to determine compliance with the mechanical properties requirements of the applicable drawing. Failure of either tension test specimen to comply with yield strength or elongation requirements shall be cause for rejection of the lot represented by the sample.

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4.3.3.2 Air pressure test.-Each metal parts assembly shall be subjected to this test to determine compliance with the requirements of the applicable drawing. Failure to comply, shall be cause for rejection of the assembly.

4.3.3.3 Joint quality.-After air test and prior to application of protective coating, two projectile assemblies from each lot shall be selected at random, sectioned and separated, to determine compliance with 3.8.6. Failure of any specimen to comply shall be cause for rejection of the lot.

4.3.3.4 Cleaning and application of protective coating.-The sample size and acceptance criteria for the inspection of cleaning the item and the application of the protective coating shall be as specified in the applicable specification, except that body assemblies will be used instead of panels. Failure to comply with the requirements of the applicable drawings and specification shall be cause for taking the corrective action recommended therein.

4.3.3.5 Magnetic particle inspection.-Heat treated stress relieved bodies after final machining shall be inspected by the magnetic particle inspection technique approved by the responsible technical agency.

a. One hundred percent magnetic particle inspection will be in effect until 2,500 consecutive items have been inspected and found free of metal defects.

b. Reduced sampling may then be instituted for magnetic particle inspection consisting of 30 bodies per hour for ten consecutive working days.

c. If no defects are found during the ten days, the sample may be reduced to 15 bodies per hour for five consecutive working days.

d. If no defects are encountered during the five days, the sample may be reduced to five bodies per hour.

e. If a defect is encountered by magnetic particle inspection during reduced sampling(b through d), the magnetic particle inspection shall revert to a above, in addition 100 percent backscreening to the last acceptable hour's production will be performed.

f. For requalification of reduced sampling, the manufacturer must follow the same preceedures as outlined in paragraphs a through d above.

g. Should a controversy arise as to the validity of an indication, the manufacturer shall use metallographic test methods or an approved nondestructive method for proof of acceptability to the responsible technical agency.

h. Continual application of a 100 percent magnetic particle inspection by paragraph a above, shall not be used as an alternate procedure in lieu of corrective manufacturing process action to eliminate a continuing process induced defect. If corrective action is not instituted and the cause eliminated, the Government reserves the right to withdrawl acceptance inspection until the contractor can prove his process is capable of producing suspect free projectiles.

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4.3.4 Functional proving ground test.-The following Proving Ground test will be performed by the Government:

4.3.4.1 First article sample.

4.3.4.1.1 Initial production sample.-The quantities listed in Table I shall be ballistically tested in the phases indicated.

TABLE I

Phase	Quantity
Excess Pressure	25
Service Pressure	25

4.3.4.2 Production lot sample.-The quantities listed in Table II shall bw ballistically tested in the phases indicated.

TABLE II

Phase	Quantity
Excess Pressure	10 each lot
*Service Pressure	10 each lot

*After three consecutive production lots have been ballistically tested and accepted for this phase, ballistic testing may be discontinued with the approval of the procuring activity. However, implementation of ballistic testing will be instituted whenever directed by the procuring activity

4.3.4.3 Excess pressure.-Samples provided in accordance with Tables I and II will be subjected to this test. Failure of any projectile assemble to comply with 3.7 shall be cause for rejection of the initial production sample or the production lot represented by the sample.

4.3.4.4. Service pressure.-Samples provided in accordance with Tables I and II will be subjected to this test. Failure of any projectile assembly to comply with 3.7 shall be cause for rejection of the initial production sample or the production lot represented by the sample.

4.3.5 Inspection Equipment.-Index of Inspection Equipment Lists, LI-10543048 identifies the inspection equipment required to perform the examination and test prescribed herein. Equipment design shall be in accordance with code designation of the applicable LI.

4.4 Test methods and procedures.

4.4.1 Mechanical properties. (See 6.3)

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4.4.1.1 Hardness.-After final heat treatment or stress relief and prior to final machining, the hardness test shall be performed on the sidewall of the body in the area of the rotating band. The test shall be performed using the equipment and procedures for performing this test as prescribed by ASTM Method E-10.

4.4.1.1.1 Bar stock lots.-After final machining, the hardness test shall be performed on the sidewall of the body in an area approximately one-half inch rear of rotating band. The test shall be performed using the equipment and procedures for performing this test as prescribed by ASTM Method E-10.

4.4.1.2 Tension Test.

4.4.1.2.1 Bodies.-From the samples selected in accordance with 4.3.3.1.2.1, 4.3.3.1.2.2, 4.3.3.1.2.3, or 4.3.3.1.2.4 tension test specimens conforming to the largest obtainable standard "R" type specimen as prescribed by ASTM Method E-8 shall be taken as follows: Two tension test specimens shall be taken longitudinally from opposite sides of the body 180 degrees apart and one transverse tension test specimen throught the base area of the body. (Centerline of the specimen shall be between .25 and .35 inches from the centerline of the base.) The number of specimens and their location is to determine the acceptability of the mechanical properties requirements and the Government shall not be restricted from testing additional specimens at other locations to determine that the mechanical properties of the projectile meet the drawing requirements. The criteria for performing this test shall be as prescribed by ASTM Method E-8, except the elongation shall be measured to the nearest thousandth (0.000) prior to conversion into percent elongation. Percent elongation shall be recorded to the nearest tenth (0.0).

4.4.1.2.2 Adapters.-From the samples selected in accordance with 4.3.3.1.2.5, one longitudinal tension test specimen conforming to the largest obtainable standard "R" type specimen as prescribed by ASTM Method E-8. The criteria for performing this test shall be as prescribed by ASTM Method E-8, except that elongation shall be measured to the nearest thousandth (0.000) prior to conversion into percent elongation. Percent elongation shall be recorded to the nearest tenth (0.0).

4.4.2 Air pressure test.-This test will be performed using equipment capable of performing the test accurately and consistently. The equipment shall be calibrated prior to the start of each day's operation and after each four hours continuous operation. The assembly shall be submerged in water in such a manner that the nose adapter joint is covered and observation can be made for the full 360 degrees of the adapter joint. The pressure will be applied until the required minimum pressure is reached after which, the pressure will be held for the minimum time specified on the applicable drawing. Observation shall be made for air bubbles rising to the surface of the water or bubbles leaving the vicinity of the adapter joint of the entire time required. After acceptance, stamp letter "A" as indicated on the applicable drawing.

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4.4.3 Joint quality.-The samples selected in accordance with 4.3.3.3 shall be sectioned and separated as follows:

a. The forward portion of the projectile assembly, 1-inch rear of the braze joint shall be removed in such a manner that no portion of the braze will be damaged.

b. The forward portion of the body containing the adapter shall be cut vertically through the center of the adapter.

c. Each portion of the body shall be peeled from the adapter and the mating pieces of the body shall be visually examined to determine compliance with the requirements.

4.4.4 Protective coating.-The procedure for testing the cleaning of components and the application of protective coating shall be in accordance with the methods and criteria for testing, as specified in the applicable specifications. The salt spray test prescribed shall be performed in compliance with ASTM Method B117.

4.4.5 Magnetic particle (Inspection method).-The procedure for performing this test utilizing the wet fluorescent method shall be as specified in MIL-I-6868, for indirect magnetization be either the continuous method or the residual method, utilizing a circular magnetic field. The suspension shall be applied to the entire body, inside and outside. After draining, the body shall be thoroughly inspected in a darkened area under an ultra-violet light which will produce adequate fluorescence of the magnetic particles. The solid content of the suspension shall be maintained at a level recommended by the suspension manufacturer. The suspension shall be checked at the start of each day's production and at intervals on not over 4 hours of continuous operation, utilizing the procedures and equipment prescribed by the specification MIL-I-6868. When performing the test forsolid content of the suspension, if it is determined that it does not comply, all projectile bodies that have been tested and accepted since the last satisfactory test was performed shall be re-tested after the suspension has been brought to the proper level. Magnetic particle inspection shall be performed on equipment that complies with the requirements of MIL- M-6867. The operator(s) of the equipment and inspector(s) making the determination shall be certified in accordance with MIL-STD-410.

4.4.6 Proving ground functional tests.

4.4.6.1 Excess pressure.-The ballistic test projectiles inert loaded and dummy fuzed will be fired in an acceptable standard gun based upon current gun tube condemnation criteria. A propellant charge adjusted to produce 109 to 117 percent of the upper pressure limit of the propellant will be used. Photographic equipment (smear camera) will be set up near the gun muzzle to record the flight of the projectiles. Visual observation will be made to assure projectile impacts in target area. Velocity and chamber pressure will be recorded.

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4.4.6.2 Service pressure.-The ballistic test projectiles inert loaded and dummy fuzed will be fired alternately with projectiles from a reference lot for comparison. The projectiles will be fired in an acceptable standard gun based upon current gun tube condemnation criteria. The projectile shall be fired for accuracy at a range selected by the proof director. Photographic equipment (smear camera) will be set up near the gun muzzle to record the flight of the projectile. Visual observation will be made to assure projectile impacts in the target area. Velocity and chamber pressure will be recorded.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.-Level C - Packing and marking shall be in accordance with Drawing F7548913.

5.2 Packing.

5.3 Body assembly.-Body assemblies shall be packed in accordance with Drawing F7548913.

5.4 Marking.-Packaged body assemblies shall be marked in accordance with MIL-STD-1169.

5.5 Shipping.-When component parts from more than one lot are shipped as a carload, each lot must be kept separated and division between lots clearly indicated to prevent mixing in transit.

5.6 Data cards.-The contractor shall prepare and furnish data cards in the number to be determined by the contracting officer in accordance with MIL-STD-1167.

6. NOTES

6.1 Visual examination qualification.-When compliance with the applicable requirement is in doubt as a result of visual examination, this characteristic may be measured or gaged to determine acceptability.

6.2 Visual standards.-Visual standards to be established at contractor's plant in compliance with 3.6.3.

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6.3 Alternate mechanical properties testing.-Alternate procedures for mechanical properties testing may be used by the contractor when such testing and procedures provide as a minimum the Quality Assurance specified herein. Prior to applying the alternate procedures, the contractor shall describe them in a written proposal and shall demonstrate for the approval of the responsible technical agency that their effectiveness is equal to or better than the Quality Assurance provisions specified herein.

6.4 Ordering data.-Procurement documents should specify the title, number, and date of this specification.

Custodian:

Army - MU

Preparing Activity:

Army - MU (FA)

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